

Governor's Honor Program Civil Engineering

DESIGN PROPOSAL
Contract GHP-CE2026-001

Preliminary Tunnel Design

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Preliminary Tunnel Design

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Contract Contact: Alex Resnick July 7, 2026
RE: GHP Civil Engineering Tunnel Design Proposal

From: GHP Civil Engineering Team
Engineering Research Building, Suite 2008
Statesboro, GA, 30460

Dear Alex Resnick, Tristin Shestag, Ben Hammond:

The GHP Civil Engineering Team is pleased to submit this proposal to provide engineering services for the Preliminary Tunnel Design. The team understands that this engagement requires an improved tunnel design to effectively replace the existing tunnel, while also incorporating modern safety codes. Additionally, the tunnel must be upsized in order to accommodate the addition of a bike lane and pedestrian sidewalks, all of which will adhere to modern standards to ensure the successful delivery and long-term reliability of this infrastructure.

Based on the RFP requirements and the subsequent project assessment, the methodology will focus on establishing an initial tunnel design that minimizes cost while providing an effective design that can replace the initial degrading tunnel. The team will focus on completing the project within the deadline while also constructing a detailed scale model (1:24 scale) for demonstration purposes. The team is also prepared to address the project's core challenge of replacing the degrading tunnel effectively without causing extensive site damage. The importance of keeping the project on schedule is recognized and will ensure strict adherence to all state and federal compliance standards applicable while keeping our proposal cost-effective.

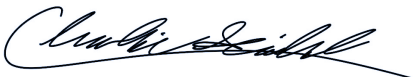
In preparation for this bid, the team acknowledges the receipt and incorporation of all revisions to the original RFP as listed below.

- Revision No. 1, received on June 28, 2026
- Revision No. 2, received on June 29, 2026
- Revision No. 3, received on July 2, 2026

The team is confident that the proactive approach to project administration, quality control, and risk management will provide the establishment with a highly efficient and successful project outcome.

The GHP Civil Engineering Team is thankful for this opportunity to present these qualifications. The team looks forward to the possibility of collaborating with the client on this important project.

Sincerely,



07/07/2026

Charlie Seidel — Date



07/07/2026

Aleena Huang — Date



07/07/2026

Peter Barahona — Date



07/07/2026

Yujin Kim — Date



07/07/2026

Madilynn Hagemes — Date



07/07/2026

Dylan Knight — Date

1. Executive Summary

A two-lane vehicle tunnel in North Georgia is in need of replacement. It was constructed in 1921, and as a host of other aging tunnel systems have experienced unexpected degradation, its replacement has been solicited to ensure long-term stability. Tunnels in mountainous regions are crucial because they provide efficient travel routes. With inaction, the currently stable tunnel may start to disintegrate, creating risk for travelers as well as increasing the difficulty and cost of later tunnel replacement.

The proposed replacement for the tunnel is a 60-foot-long horseshoe tunnel, maintaining the geometric shape of the 1921 structure while expanding the overall tunnel to accommodate multi-modal traffic in strict compliance with applicable federal/state laws and regulations. The updated infrastructure includes two, 12-foot vehicle lanes sized for large commercial vehicles, a 10-foot at-grade bidirectional bike lane isolated from vehicular traffic by a 2-foot safety buffer zone, and a raised 8-foot pedestrian walkway fully conforming with ADA regulations and protected by a structural safety guardrail. To withstand the seismic zone's 0.075 g ground acceleration, calculations determined the necessary tunnel thickness of 1 foot, which will be reinforced with steel fibers mixed into the concrete and an internal rebar cage. The tunnel will be lit by a 12-lamp grid with 6 fixtures mounted 10 feet high at 10-foot intervals along each interior side. Water management relies on a drainage system that utilizes trench drains and catch basins. The trench drains will run along a 3-foot concrete walkway margin on the left side of the roadway alongside precast catch basins placed at the pedestrian walkway transitions at both tunnel exits. These surface systems connect under the road and then feed a centralized wet well rigged with an automatic primary submersible pump. Calculations were done to determine ventilation needs and established that natural airflow is sufficient and no mechanical equipment is required. System utilities will be regulated by an external electrical hut and distribution box outside the tunnel, which houses the primary control grid and critical safeguards, including an auxiliary secondary pump and backup generator.

The GHP Civil Engineering Team is presenting a detailed Preliminary Tunnel Design proposal that balances both user and client satisfaction. The team took a subsystem approach to the problem, dividing the requirements into their respective subsystems, including electrical, ventilation, drainage, structural, and roadway designs. The electrical aspects of the tunnel include addressing light placement, power sourcing, connection to other subsystems, and wiring. Ventilation requirements are determined by the tunnel's ability to dilute contaminants through natural airflow as discussed in Appendix C. The tunnel is designed to accommodate the expected rainwater entering the tunnel, providing a safe way to control the release of rainwater and ensuring that it prevents the accumulation of stagnant water in the roadway by the drainage system. Structural designs focus on the tunnel structure, ensuring it withstands its own weight, the overburden loads of the mountain, and the stress induced by minor seismic activities. The roadway design is primarily focused on addressing the needs of vehicles, pedestrians, and cyclists.

The GHP Civil Engineering team will present their ideas, thought processes, and overall designs in the following deliverables, further expanded upon in the quality assurances outlined in Section 4.

- Tunnel Design Floorplan

- Tunnel Elevation Drawing
- Isometric Design Drawing of Piping System
- P&ID and One Line Diagrams
- Utilized Calculation Documents
- Bill of Materials
- Code of Record

The estimated overall project investment is $5,198,000 for an anticipated 12-month construction time period. This comprehensive total brings $1,464,000 in preconstruction costs and a base construction phase cost of $3,734,000. This 12-month budget is delivered under a fixed-price turnkey contract to ensure strict cost certainty and eliminate public funding overruns. Every figure is grounded in the master Bill of Materials, balancing all materials, specialized field crews, and necessary heavy machinery assets. By finding the numbers on localized engineering benchmarks and optimizing deployment to eliminate unnecessary asset idle time, The GHP Civil Engineering Team ensures this complete estimate fully covers all structural elements required by the RFP.

The growing risk of structural degradation along this critical mountain route makes fixing the tunnel an urgent safety and economic priority. The selection committee is invited to partner with the GHP Civil Engineering Team to turn this aging 1921 structure into a safe and durable asset for North Georgia. Please review the attached floor plans, calculation files, and project deliverables, and contact the project manager today to formalize this proposal and secure an active 2026 mobilization window.

2. Organization and Approach

The following is the GHP Civil Engineering Team organization and approach to meeting the objectives of the project:

2.1. Roles and Organization of the GHP Civil Engineering Team

Project Staff and Team Experience

There are six official members of the GHP Civil Engineering Team, each spanning various roles pertaining to the project (outlined in Section 2.4).

Each team member of the GHP Civil Engineering Team brings substantial experience in construction-related projects, including work in plumbing systems, greenhouse design, hydroponics systems, and other engineering applications. However, this project marks the team's first opportunity to collaborate on the analysis and preliminary design of a mountainous tunnel, introducing a specialized area of civil engineering beyond their previous experience. Despite entering a new discipline, the team's strong foundation in construction methods, cost estimation, structural systems, and project planning has been instrumental in the successful completion of the Preliminary Tunnel Design.

2.2. Facilities Available

The facilities available for the use of the GHP Civil Engineering Team include:

- The Georgia Southern Engineering Research Facility, Rooms 1055 and 2008
- The Georgia Southern Zach S. Henderson Library

2.3. Project Management Approach

The GHP Civil Engineering Team is organized into many different roles, as outlined in Section 2.4. Overall, the scope of the Preliminary Tunnel Design is to create an official proposal and a 1:24 scale model of the tunnel design. This includes subsystems within the scale model, such as the electrical and drainage systems. The team shall approach the design with a top-down method, starting off with research pertaining to tunnel shape, overall structure, materials, and more.

After this research shall be completed, the team will break off into their respective subteams (including calculations for wall thickness, seismic considerations, electrical systems, etc.) in order to maximize time and efficiency.

Each section of the proposal must be assured via the Quality Assurance Review Template, as found in Section 4.3.2.

2.4. Roles of Key Individuals on the Team

All team members listed in Table 1 contributed to the planning of the construction of the tunnel, including mold creation, concrete work, sanding, etc.

Table 1 - List of Key Team Members, Roles, and Role Description

Name	Role	Description
Aleena Huang	Structural Engineer	Responsible engineer for developing structural calculations, and optimizing wall thickness related to tunnel pressure and external factors.
Dylan Knight	Structural Engineer	Responsible engineer for the structural tunnel shapes and designs, analyzing key factors in the tunnel design.
Charlie Seidel	Transportation Engineer	Responsible engineer for the design of transportation systems, including roadways, sidewalks, and bicycle lanes.
Peter Barahona	Ventilation Engineer	Responsible engineer for calculating ventilation requirements, seismic loads, and safety rules pertaining to future sustainability of the tunnel.
Madilynn Hagemes	Drainage Engineer	Responsible engineer for plumbing and drainage systems, including the schematic and the creation itself.
Yujin Kim	Electrical Engineer	Responsible engineer for the electrical systems design, including electrical schematics, auxiliary power, and lighting systems.

3. Detailed Design Description

3.1. Code of Record

The design adheres to state and federal codes. Specific codes for each element of design are listed below.

3.1.1. Structural

- Technical Manual for Design and Construction of Road Tunnel - Civil Elements, FHWA-NHI-10-034, First Edition, Federal Highway Administration, 2009
- AASHTO LRFD Road Tunnel Design and Construction Guide Specifications, First Edition, American Association of State Highway and Transportation Officials, 2017
- Standard Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement, ASTM A615/A615M-26, ASTM International, 2026, ASTM A615/A615M-26 Standard
- Engineering and Design: Tunnels and Shafts in Rock, EM 1110-2-2901, Current Edition, United States Army Corps of Engineers, 1997

3.1.2. Roads

- LRFD Road Tunnel Design and Construction Guide Specifications, 1st Edition, AASHTO, 2017
- Guide for the Development of Bicycle Facilities, 5th Edition, AASHTO, 2024
- A Policy on Geometric Design of Highways and Streets (Green Book), 8th Edition, AASHTO, 2018
- Bikeway Selection Guide, 1st Edition, US Department of Transportation, 2019
- Accessibility Standards, Americans with Disabilities Act (ADA), 2010
- Design Policy Manual, Revision 7.5, Georgia Department of Transportation, 2026

3.1.3. Electrical

- Daytime Lighting in Short Tunnels, Final Report, GDOT Research Project 22-31, Georgia Department of Transportation, 2023
- Maryland SHA Bicycle and Pedestrian Design Guidelines, Illuminating Engineering Society of North America, 2012

3.1.4. Drainage

- 2018 International Plumbing Code (IPC), IPC-2018, International Code Council (ICC), 2017
- Drainage Design Policy Manual (DDPM), Georgia Department of Transportation (GDOT) - Office of Design Policy and Support, Revised 11-25-2024, November 25, 2024

3.1.5. Ventilation

- Road Tunnels: Vehicle Emission and Air Demand for Ventilation, 2nd Edition, PIARC, 2019
- Standard for Road Tunnels, Bridges, and Other Limited Access Highways, 2nd Edition, NFPA 502, 2026

3.1.6. Permitting for the Overall Tunnel

Before tunnel construction, the design teams must attain all proper permitting and licensing about the construction of the tunnel, including but not limited to: National Pollution Discharge Elimination Permitting requirements, such as wasteload allocation request, anti-degradation analysis, EID, CWSRF projects, DDR, P&S, and watershed assessment and protection plans, as well as the Traffic Interruption Request packet requirements, including the TIR packet and proof of Temporary traffic control plans, in addition to all further applicable permitting required by epd.georgia.gov

3.2. List of Design Calculations

All calculation requirements pertaining to design choices are calculated and provided within the respective figures of Appendixes as displayed with the following:

- Structural Calculations that pertain to the stress induced on the tunnel and thickness of wall lining (Appendix A)
- Seismic Calculations relating to the stress induced from ground acceleration affecting wall thickness (Appendix B)
- Ventilation Calculations displaying justification for sufficient natural ventilation without added mechanical ventilation (Appendix C)
- Electrical Calculations depicting electrical needs for all systems within the tunnel (Appendix D)
- Drainage Calculations regarding pipe sizing and placement to prevent stagnant water build up within tunnel (Appendix E)

3.3. List of Design Drawings

Preliminary design drawings displaying key features are provided within the respective figures of Appendix F as displayed with the following:

- Floorplan showcasing key design features in road lanes, pedestrian side walks, and bike lanes (Appendix F Figure 3)
- Plumbing and Instrumentation Diagram depicting all plumbing and electrical systems within the tunnel and their interaction (Appendix F Figure 4)
- Elevation showing depth of all systems within the tunnel (Appendix F Figure 5)
- Electrical One-line showcasing the flow of electricity and all required systems within the tunnel (Appendix F Figure 6)
- Isometric showcasing three dimensional depiction of all drainage and electrical systems within the tunnel (Appendix F Figure 7)

3.4. Tunnel Design - Horseshoe Shape

The group will design the preliminary tunnel using a horseshoe-shaped cross section after analyzing alternative tunnel structures such as circular and rectangular sections. The horseshoe structure was considered the most optimal design as it provides high structural stability, constructability, and long-term functional conditions. According to the Reference Guide for Load Rating of Tunnel Structures Section 2.2.1.3, “Positively, a horseshoe

"provides a concentric circular load path, which helps distribute loads through compression around the main liner system." Pertaining to the geography of the Preliminary Tunnel Design, horseshoe-shaped sections are ideal because the Design Equations for Predicting Stability of Unlined Horseshoe Tunnels in Rock Masses states that "The uses of horseshoe-shaped sections are not uncommon in the construction of tunnels and subways, particularly in mountainous areas where rock excavation is needed.

All costs pertaining to the tunnel design, including material, equipment rental, and labor usage, are categorized and provided in Appendix G table 5. Costs remain subject to change throughout the duration of construction, with minor time accounting for weather delays or accidents; all calculated costs should be used as a baseline for project construction. Cost determined relevant to the date 7/7/2026, delay of proposal acceptance or notice to proceed may be subjected to future economic factors.

3.5. Potential Project Challenges

3.5.1. Critical Design Challenges

Stress concentration is a critical design issue for the horseshoe tunnel arch, as the arch points that connect with the straight edges of the exterior tunnel shape become fragile and susceptible to fracturing. However, according to the Structural Calculations (Appendix A), as The GHP Civil Engineering Team is assuming the tunnel hole consists of hard, granite-like rock, the structure is able to sustain itself. When taking the seismic considerations into account, the wall thickness of eight inches is enough to stabilize the tunnel at the point of an arch, while the wall thickness of one foot is generously above the minimum to sustain the tunnel in the event of an earthquake. Therefore, under normal circumstances, the stress concentration of the tunnel should not be a significant issue. However, in the event of greater seismic activity or expected pressure, footings can be placed on the interior of the tunnel walls. These footings shall slope inwards from the tunnel walls to the tunnel floor, creating greater stabilization.

Emergency exits could be considered a critical design issue for the Preliminary Tunnel Design; however, according to the Ventilation Calculations (Appendix C), because the tunnel is only 60 ft long, emergency fires due to ventilation issues do not need to be accounted for. This means that these effects can be considered negligible. Refer to Appendix C for greater details.

Although there are no explicit design concerns pertaining to electrical and ventilation systems, even if new issues are brought to light in the future, many disasters are naturally mitigated due to the short length of the tunnel. As the tunnel is only 60 feet long, electrical and ventilation systems are often unnecessary. This is seen in precedents via *The Georgia Department of Transportation (Daytime Lighting in Short Tunnels)* which states, "...It is commonly assumed that natural sunlight can sufficiently illuminate [short tunnels <410 ft], obviating the need for artificial daytime lighting." and *Natural Ventilation of a Short Road Tunnel* which demonstrates in Table 1 that existing tunnels around the world that are <547 ft are all illuminated by natural lighting.

3.5.2. Environmental Challenges

Earthquake resistance is referenced by the Seismic Calculations (Appendix B), at a magnitude of 0.075 gram of ground acceleration, it is standard for a wall thickness of 12 inches to sustain this pressure. According to the Structural Calculations (Appendix A), the minimum pressure that high-strength concrete must withstand is 4.0 ksi, which means that this wall thickness is sufficient to address seismic concerns. Drainage Calculations (Appendix E) addresses flood mitigation methods through the drainage systems.

Additionally, a downstream water assessment will need to be completed based on Section 2.2.2 of the GDOT Stormwater Design Guide.

of the Georgia Stormwater Design Guide. As this area is a non-MS4 area, The GHP Civil Engineering Team must design a report that is sufficient to convince the GDOT that detention is not necessary for implementation. Following the GDOT Standard 2.2.3, an additional addendum will need to be submitted that details construction plans and outflow areas. This report will require us to determine peak flow rates and environmental impacts, possibly calling for the obtaining of a BMP.

3.5.3. Cost and Schedule Efficiency

Cost and schedule could be minimized with a few different methods:

- Gantt Charts are highly effective in minimizing scheduling and timeline, as they provide a clear and concise visual timeline of the Preliminary Tunnel Design. Additionally, the Gantt Chart of the Schedule of Work (Section 3.6) provides sufficient allotted amounts of time for each task, while overlapping certain tasks in order to maximize efficiency. Gantt Charts help estimate to organize the date that adapts to the weather delays.
- Additionally, via the Schedule of Work (Section 3.6), heavy machinery is only rented during the weeks that each piece of equipment is deemed necessary while the number of weeks needed are condensed in order to save money on heavy machinery transportation. However, extra time is still built into rental periods for equipment in case of delays.
- The cost is minimized by allocating a sufficient block of money for the budget, and having a general idea of how much money each item costs before the construction of the tunnel. Furthermore, a contingency budget is set aside for emergency funds.

3.6. Schedule of Work

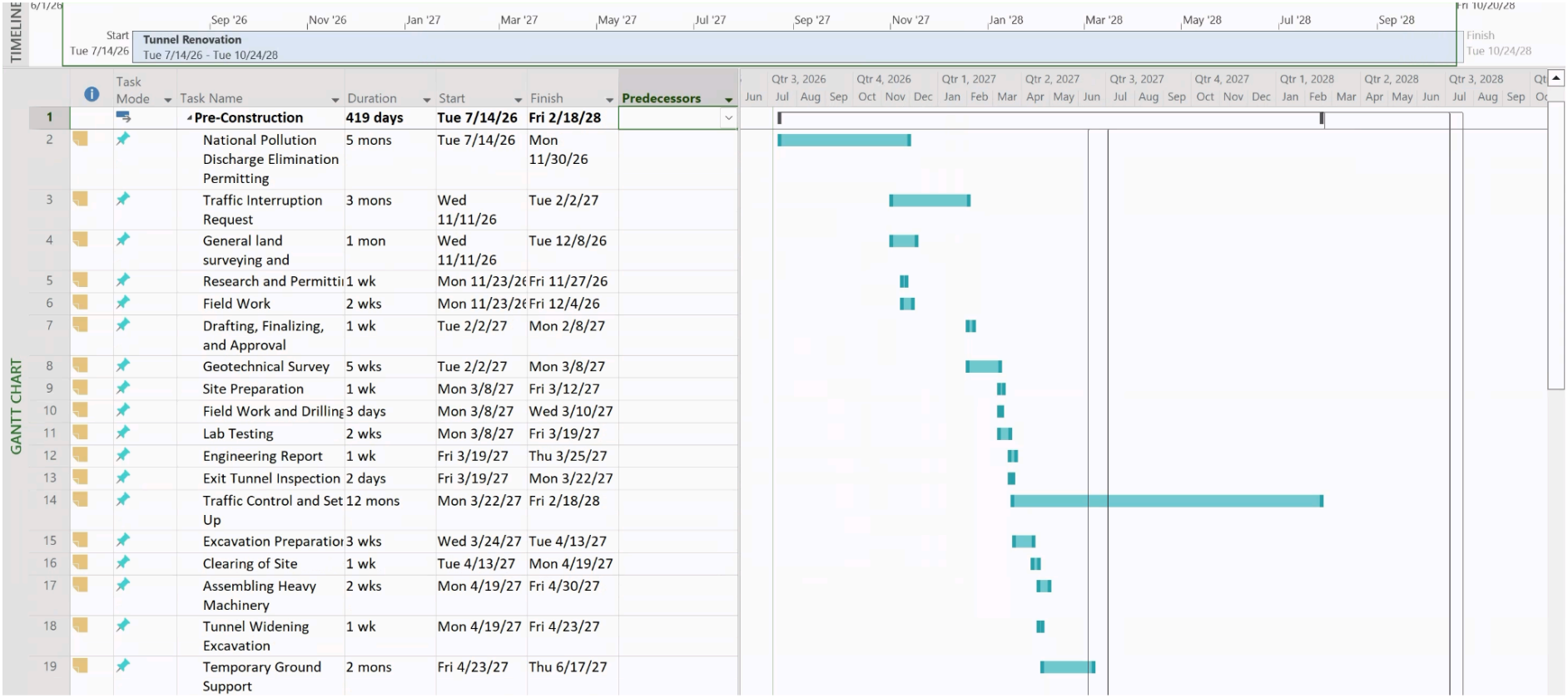


Figure 1 - Gantt Chart Related to the Schedule of Work



GANTT CHART

20	**Structural**	**278 days**	**Mon 6/21/27**	**Wed 7/12/28**	**1**
21	Insert Steel Lining Reinforcement Pages	6 days	Mon 6/21/27	Mon 6/28/27	
22	Concrete Formwork	2 wks	Thu 6/29/28	Wed 7/12/28	
23	**Roads and Sidewalks**	**266 days**	**Mon 7/12/27**	**Mon 7/17/28**	**1**
24	Basic Road Constructic	1 wk	Mon 7/12/27	Fri 7/16/27	
25	Sidewalk Construction	2 days	Sun 7/16/28	Mon 7/17/28	
26	Road Painting	1 day	Mon 7/19/27	Mon 7/19/27	
27	**Electrical**	**40 days**	**Mon 7/10/28**	**Fri 9/1/28**	**1**
28	Pre-Installation Planni	2 wks	Mon 7/10/28	Fri 7/21/28	
29	Check with Codes and Standards	2 wks	Mon 7/10/28	Fri 7/21/28	
30	Electrical Coordination and	1 wk	Mon 7/24/28	Fri 7/28/28	
31	Material	3 wks	Mon 7/24/28	Fri 8/11/28	
32	Commissioning	3 days	Fri 8/11/28	Tue 8/15/28	
33	Installation & Testing	2 wks	Tue 8/15/28	Mon 8/28/28	
34	Final Approval	1 wk	Mon 8/28/28	Fri 9/1/28	
35	Project Closeout	1 wk	Mon 8/28/28	Fri 9/1/28	
36	**Drainage**	**67 days**	**Mon 7/24/28**	**Tue 10/24/28**	**1**
37	Plumbing Grading/Drill	1 mon	Mon 7/24/28	Fri 8/18/28	
38	Concrete Work	2 wks	Mon 8/21/28	Fri 9/1/28	
39	Vault and Vent/Basin Installation	2 wks	Mon 9/4/28	Fri 9/15/28	
40	Piping and Storm Main Connection	2 wks	Mon 9/11/28	Fri 9/22/28	
41	Electrical and Valve Finishes	2 wks	Mon 9/25/28	Fri 10/6/28	

Figure 2 - Gantt Chart Related to the Schedule of Work (cont.)

4. Project Specific Quality Plan

Defines the systematic quality assurance and quality control processes used to ensure all deliverables for the Preliminary Tunnel Design are executed with technical accuracy and interdisciplinary alignment.

4.1. Purpose of Quality Management Plan

The purpose of the Quality Plan is to detail how the preparation and review processes for the **GHP 63 Civil Engineers** will be implemented to ensure that the project outputs are delivered with a high degree of technical accuracy. This will be achieved by ensuring that all project management processes are conducted with quality (quality assurance) and by developing criteria for the outputs themselves (quality control).

This document is to be read in conjunction with the **Preliminary Tunnel Design** and will be reviewed and amended to meet new conditions or objectives during the project's life span.

4.2. Quality Management Plan and Components

To achieve this, the **Preliminary Tunnel Design** Quality Management Plan will include the following components:

Quality Assurance - ensures quality project management processes, including a distinct understanding of the team's goal, a clear setup of deliverable deadlines, and a line of communication between workers to ensure system compatibility. All communication outside of working hours between the project preparers and project managers, as well as comments from checkers or reviewers, must be documented in writing to uphold a clear record of communication.

Quality Control - via the development of quality outputs, ensuring specific methodologies and standards for the GHP Civil Engineering Team to follow, ensuring the utmost quality of work. Each quality review consists of a preparer, checker/reviewer, and approver, with roles further detailed in the project calculations section (Appendices A - E). The preparer creates the work and should not be involved in the checking, reviewing, and approving process. The main objective of the checker/reviewer is to evaluate proper formatting and technical errors. The main objective of the approver is to grant final approval and sign off on the document before the next stage of the project can be initiated.

Quality Improvement - Once passed through the check or review, errors identified will be documented and returned to the preparer. Specific questions the checker/reviewer is able to utilize are detailed in this section. Overall, this section outlines all steps necessary for quality improvement; see Section 4.4 for more details.

Deliverables - Detailed drawings and schematics relating to the project, including, but not limited to, a floor plan, elevation map, P&ID, One-Line Electrical, and Isometric drawing. Additionally, the project report will include all required calculations detailed in Appendices A through E that define the structural, seismic/shear strain, ventilation, electrical, and drainage necessities for the project.

4.3. Project Quality Control

4.3.1. Methodologies and Standards

Quality Assurance for the **Preliminary Tunnel Design** will be achieved with relevant quality management processes.

4.3.2. Quality Review

Table 2 - Preliminary Planning and Design Descriptions

Description	Comments
Project Calculations	Reviews will be conducted following the calculation proposals and will be made up of one or two individuals. Each quality review will consist of a checker and a reviewer. At the end of each check/review/approval, the individual responsible for the quality review will sign their name at the end, signifying a successful review. Each check will cover: - Proper use of formatting in accordance with the **Calculation Template** - Logical flow of thinking - Rational use of assumptions - Verify calculations Each reviewer will cover: - Evaluate the project from a broader interdisciplinary perspective - Ensure design integrates smoothly with other disciplines - Verify design meets client specifications, project goals, and industry specifications Each approver will cover: - Final act of approving/sign off - Moving the item/procedure to the next stage in its lifestyle - Official/operational consent to conclude that specific stage of the project During the calculation processes, a team/individual will prepare the needed calculations, a second team/individual will check and review the calculations, and a final team/individual will approve the calculations. Errors found during the checking and reviewing phase will be returned to the preparing

	team for the needed adjustments to calculations.
Management of Changes to Project Scope	It's important to anticipate and plan for possible changes through risk contingency plans: - Keep track of emerging or unanticipated issues through issue management procedures - Procedures include firm identification and assessment of issues before preparing potential solutions as a team. - Bring issues, which could have a major impact on the nature or substance of the project, to the relevant team members of the group as a whole and project managers, so they can re-evaluate the project or make adjustments. - Use an interactive process of change within the scope of work of a single project. - Conduct impact analysis to evaluate how potential solutions affect the project baseline, and implement change once approved by the project team and stakeholders. - Reflect changes of project scope in Preliminary Tunnel iterations - Keep track of the updates and revise failed concepts. After identifying the problem, implement the change needed after all revisions are approved.

4.3.3. Information Management

Table 3 - Document Organization and Management Standards

Description	Comments
Document Management	Acceptance criteria are outlined in the contract. Version control will be managed via Slack, Google Workspace, and Microsoft 365. Copies of versions will be stored via the drive, and every change and revision will be logged. Organization is ensured by naming documents by [date][document/project][description][version]. For questions, refer to Alex Resnick, Tristin Shestag, or Ben Hammond via Slack.
Record Keeping	All official documents relevant to the proposal will be

	documented in both the official proposal document and Slack.

4.4. Quality Improvement

4.4.1. Output Quality Criteria

Ensure that any materials, products, processes, and services required for output development are fit for their purpose. The quality improvement process is outlined below:

- See 4.3.2 for review process: Once the checker and reviewer provide feedback for a component of the project, the work is then returned to the preparer.
- The preparer will evaluate all comments, ideas, and questions, and edit/change their work accordingly.
- Once the work is edited by the preparer, it should then be sent back to the checker/reviewer in a timely fashion for another round of checking/reviewing.
- Repeat this process as seen fit.
- Once the checker/reviewer signs off, the work should be passed on to the approver, who will pass on the work to the next stage of its lifecycle.

4.4.2. Quality Control Measurement Methods

- A series of questions can be utilized per the checker/review to determine if further changes/edits are needed:
- These questions should serve as a guide and shall vary accordingly depending on the work or the specific part of the project.
    - Does this process comply with all industry standards?
    - Can this process realistically be utilized in the real world? If not, what are the nuances associated with it?
    - Does this work seem feasible for the scale model/project of the Preliminary Tunnel Design? If not, what should be changed?
    - If the output fails a criterion, what is the protocol to resolve its flaw?
    - Have all non-destructive and functional tests been completed?
    - Are the final dimensional tolerances within allowable limits?

4.5. Quality Components and Deliverables

4.5.1. Drawings and Schematics

Drawings and schematics included in the design will pertain to the overall tunnel structure as well as the integrated systems that must be a part of the tunnel design. The floor plan will provide a detailed overhead view of the tunnel structure itself, while also including basic markings of primary system components. The elevation drawing will be completed similarly, but will provide a side view of the tunnel and possibly cross-section schematics if needed. To further detail the systems integrated in the tunnel design, a P&ID drawing and a One-Line Electrical drawing will also be drafted. The P&ID will provide a 2D overview of the plumbing and drainage systems in the tunnel, while an additional Isometric drawing (3D) will provide additional information about the water systems. The One-Line Electrical drawing will be the primary document that shows the electrical systems in the tunnel, pertaining to both powering the drainage and ventilation systems and also the light system in the tunnel. A separate drawing will detail the light system

independently, while an additional schematic will be used for the ventilation system if needed.

4.5.2. Required Calculations

The overall design must include, but is not limited to, calculations regarding **drainage**: the expected amount and flow rate of water entering the tunnel, the required pipe sizing and energy (i.e energy and pressure tracking of water throughout the system), and the required pump power to effectively manage floodwaters in the tunnel; **ventilation**: the expected exhaust buildup from traffic, the required airflow to ensure safe traveling environments, as well as energy requirements to power such fans and ventilation systems; **electrical**: calculations outlining the required amount of light sources to provide sufficient lighting in terms of lumens, energy requirements for lighting systems, total energy requirements in conjunction with the drainage and ventilation systems, and documentation for the energy source(s) required to power all tunnel systems; **Seismic and Structural**: calculations defining the stress and strain placed on the project both due to ground acceleration and weight of the mountain, as well as required tunnel thickness to support the previously calculated strain placed on tunnel. All preceding calculations must be done with respect to the official tunnel sizing and be in accordance with the Project Calculations (Section 4.3.2) of the Quality Review.

4.5.3. Auxiliary Documents

Document must include a detailed Bill of Materials identifying all required resources in detail while also factoring in all costs associated with the project, including resources, labor, and general construction costs.

A clear record of all codes and standards pertaining to every aspect of the tunnel must be cited to ensure the tunnel's compatibility with state and federal requirements.

Each aspect above must be immediately recorded when completed in the respective location to ensure reliability and quality, while ensuring every aspect of the project is recorded.

Appendix A - Structural Calculations

The calculations provided in this Appendix display the justification and requirements for key design features.

1. Structural Calculations

1.1. Purpose

The purpose of the calculation is to guarantee that the wall thickness of the tunnel meets all necessary constraints and ensures safety for all civilians and vehicles passing through the tunnel. Additionally, stress, strain, and bending moment, which were solved via the Structural Calculations in Appendix A, can be referenced in all future endeavors pertaining to the maintenance and upkeep of the tunnel.

1.2. Summary of Results

In order to maintain proper structural integrity of the Preliminary Tunnel Design, the wall thickness must be **12 inches thick.** This was calculated via the Geometric Ratio Formula, which is usually relevant when sizing pipes; however, due to the irregular horseshoe shape and the lack of wall pressure of the tunnel, the expected thickness of the "circular pipe" was then manipulated to be enlarged, accounting for the weaker walls of the horseshoe shape. Therefore, this calculation successfully accounts for the external factors and parameters of the tunnel wall thickness.

1.3. Input Requirements

The following inputs were used in support of this calculation:

Geometric Ratio Thickness: ($T = \frac{Perimeter}{2\pi \times 20}$) Visual representation given in Segmental Lining Thickness vs. Internal Diameter Ratio, A Perspective Approach for Lining Dimensioning (Figure 3)

- Basis: The perimeter divided by 2PI is the estimated radius of the entire structure (assuming a circle in this scenario)
- Multiplying the radius by 20 is a standard number of thickness reinforcement for a medium-short length tunnel, as proved by Designing a Tunnel by Spirros Massinas
- The perimeter of the cross section is derived from Onshape (85.398 ft)
- The conversion from feet to inches is (*12 in/ft)
- The multiplying factor of the thickness is 1.5 (see Assumptions in Section 1.5) as the joints of the tunnel are naturally weaker than the arched roof)

1.4. Assumptions

- Assume that the mountain (composed of hard granite rock) is able to sustain itself without any concrete reinforcements in the tunnel hole.
- The hole can sufficiently maintain its structure without a concrete tunnel.
- The tunnel is leveraged to withstand seismic loads.
- Other loads are not to be considered, as groundwater does not exist due to low aquifer depth.
- It is assumed the joints of the horseshoe tunnel are weaker than the arched roof, and must be structurally accounted for in terms of wall thickness.

1.5. Calculations

- $T = \frac{85.398\ ft}{2\pi \times 20}$
  - The perimeter of the tunnel was plugged into the equation of geometric ratio thickness, ($Perimeter/\ 2\pi\ \times\ 20$) delivering the answer of (0.679922163 ft) thick.
- $0.679\ ft\ \times\ 12\ in/ft$
- This was then multiplied by (*12 in/ft) in order to convert feet to inches.
- $8.159\ in\ \times\ 1.5$
  - Finally, the value of (8.159 in) was multiplied by a factor of 1.5 to account for the structural instability of the joints of the tunnel, erring on the safer side to ensure structural stability.
- The official thickness of the tunnel is ~ **12 in**

1.5.1. Software Usage

- Excel calculations were utilized in reference to equations above. Additionally, Onshape was referenced in order to correctly calculate the perimeter of the tunnel opening.

1.6. Referenced Codes and Standards

Tunnel Shape (Horseshoe Bend) - Experimental investigation of mechanical behavior of horseshoe-shaped segmental tunnel linings
Tunnel Height (16 ft+) - LFRD 2.7.1

Industry Standard Tunnel Wall Thickness - For concrete placed with a slick-line, the minimum practical lining thickness is about 230 mm (9 in), but most linings, however, require a thickness of 300 mm (12 in) or more. - US Army Corps of Engineers (page 174)

1.7. Structural Calculations for Extreme Pressure

1.7.1. Purpose

These structural calculations should be accounted for in events of extreme pressure of granite rock placed upon the concrete tunnel. For this purpose, these should not be used officially–however, they serve an important role in demonstrating an unideal scenario in the environment.

1.7.2. Summary of Results

In an unideal scenario of rock pressure, the tunnel wall thickness of about **3 ft** is deemed necessary for the structural stability of the Preliminary Tunnel Design. This number was understood through multiple calculations.

1.7.3. Input Requirements

- Stress - ASTM C39/C39M
- Strain - ASTM D 3148, page 3
- Bending Moment - Hibbler Structural Analysis 9th Edition, page 626
- Wall Thickness - ASTM F1216

- Young's Modulus (of Reinforced Concrete) - Young's Modulus of Elasticity Common Values
- Reinforced Steel (60 ksi+ Yield Strength) - ASTM 615
- Steel Mesh Wire - Industry Standard (11x11 in spacing) Industry Standard Mesh
- Concrete Minimum (4.0 ksi) - AASHTO LRFD, Section 5
- The tunnel length (60 ft), the diameter (48 ft), and the height (29 ft), and a wall height of 5 ft.
- Material limitations were noted by industry-standard materials such as concrete strength (3.5 ksi/504000 psf) and steel minimums (60 ksi).
- The rock density was estimated (170 lbs/ft^3), the total mountain height is (80 ft), and the resulting overbearing pressure (13600 psf) was calculated by multiplying rock density and mountain height.
- The Young's Modulus (E=62640000000 Pa) was derived via concrete standards
- The Dynamic Stress (320766663 lbs)
- The tunnel was modeled in Onshape, which specified the following values:
    - The precise surface area of the internal roof (A_Roof=4523.89 ft^2)
    - The precise cross-sectional area (A_Cross=828.3 ft^2)
    - The precise perimeter of the cross-section (Pe_Cross=85.398 ft)

1.7.4. Assumptions

Assume that the mountain applies uniform pressure from all directions. There is no groundwater consideration because there is low aquifer depth. Assume that local soil composition samples of North Georgia mountain regions apply, giving an approximate soil density of 170 lb/ft^3.

1.7.5. Calculations

- $Stress = \frac{F_{all}}{A_{all}}$
    - The stress must be calculated in order to understand how much force is applied to the structure itself, and how it is able to sustain itself (74874 psf).
- $A_{wall} = H_{wall} \times L_{wall}$
    - The walls themselves (not including the arch) are solved via this formula.
- $A_{all} = A_{wall} + A_{wall} + A_{roof}$
    - Once all of the areas are added together, the surface area of the entire tunnel can be solved via this formula.
- $F_{all} = (P_{all} \times A_{roof}) + Dynamic\ Stress$
    - This includes the force of the tunnel itself, the rock, and the dynamic stress.
- $Strain = \frac{Stress}{E}$
    - Young's Modulus (E) was solved via estimation/industry standards, and this is the measurement of what is being pulled apart.
- $P_{all} = P_{overbearing} + P_{ConcreteSelf}$

    - The total pressure of the tunnel includes both the pressure of the tunnel itself, and the overbearing.
- $A_{Inside} = A_{cross} \times length$
    - The entire interior volume can be calculated through multiplying the area of the cross section by the length of the tunnel.
- $Bending\ Moment = \frac{wD^2}{8}$
    - This is the standard bending moment formula for a tunnel arch shape, and the bending moment should be utilized to understand the tunnel's curvature in response to the pressure. Bending moment = 4003200 psf
- $A_{required} = \frac{F_{all}}{F_{concrete}}$
    - This should be utilized to understand how much area needs to be required to sustain these immense amounts of force placed upon the tunnel.
- $Wall\ Thickness = \frac{A_{required}}{Per\ Cross}$
    - This is the standard calculation for a pipe, and by dividing the A_Required by the Per_Cross, it can be seen how thick the structure must be in order for it to sustain itself.

Key Takeaways:
Although an abundance of values were solved for (and can be used for future calculations and references, most critical calculations for wall thickness include:

See equations above for specific formulas
- Stress = 74874 psf
- Strain = 1.195E-06 psf
- Bending Moment = 4003200 psf
- Area Required = 282.683 ft^2
- **Wall Thickness = 3.310 ft**

1.7.6. Referenced Codes and Standards

Tunnel Shape (Horseshoe Bend) - Experimental investigation of mechanical behavior of horseshoe-shaped segmental tunnel linings

Tunnel Height (16 ft+) - LFRD 2.7.1

Appendix B - Seismic/Shear Strain Calculations

The Calculations provided in this Appendix display the justification and requirements for key design features.

1. Seismic/Shear Strain (Dynamic Stress) Calculation

1.1. Purpose

Shear strain accounts for the dynamic stress applied to the tunnel during an earthquake or other seismic event. At the given site, the tunnel is subjected to a ground acceleration of 0.075 g, which is not negligible for the tunnel design. In the case of an unforeseen seismic event, the minimum wall thickness of the tunnel must be strong enough to withstand the given acceleration on top of the static stress in order to prevent a collapse of the tunnel. This calculation is important to determine the amount of concrete needed and the minimum wall thickness of the central tunnel structure.

1.2. Summary of Results

The results from the calculations below show that the maximum tolerance of the tunnel due to seismic stress is 0.07 ksf. This result is to be added into the static stress calculations to determine the overall stress capacity of the structure, to then be utilized in the calculation of overall wall thickness. This result assumes the structure is being surrounded by free-standing rock and sediment pressing down on the structure, and may not be applicable in cases of self-sustaining rock such as bedrock

1.3. Design Inputs

Peak Ground Acceleration - 0.075 g
Density of mountain Soil - 0.17 ksf
Length of tunnel - 60 ft
Depth - 80 ft
Width - 50 ft

1.4. Assumptions

For the given site, assume the mountain applies uniform pressure from all directions. There is no groundwater consideration because there is low aquifer depth. Assume that local soil composition samples of North Georgia mountain regions apply, giving an approximate soil density of 170 lb/sf.

1.5. Calculations

Dynamic stress is located in equation 1.2; the required variable is sigma_v, which is calculated by equation 1.1 . For equation 1.1, using the given values, the density of soil (170 lb/sf) multiplied by the sum of the depth of the tunnel and height of the overall tunnel (80 ft, 29ft, respectively) to calculate the vertical overbearing.
Equations used:
$\gamma_t \times (H + D) = \sigma_v$ (Eq. 1.1)

Step 2 - Using the vertical overbearing equation 4.2.2.B, the values 0.075 g (ground acceleration), 9.8 m/s^2 (force of gravity), and the value 0.5 (stress reduction factor based

on No. FHWA-NHI-10-034 (2009) Chapter 13-6) to calculate that the maximum dynamic stress is 0.07 ksf.
Equations used:
$\frac{PGA}{g} \times \sigma_v \times R_d = \tau_{max}$ (Eq. 1.2)

1.6. Software Usage: Calculations were carried out using Microsoft Excel

1.7. Referenced Sources, Codes and Standards
Federal Highway Administration Publication No. FHWA-NHI-10-034 (2009) Chapter 13 were used to define the equations used for shear stress and total vertical overburden.

Appendix C - Ventilation Calculations

The Calculations provided in this Appendix display the justification and requirements for key design features.

1. Ventilation Requirements

1.1. Purpose

The project proposal dictates that the tunnel shall have ventilation requirements considered in the design of the tunnel. Thus, according to NFPA 502 11.1.1, for tunnels shorter than 3280 ft, emergency ventilation is not required, assuming the tunnel generates enough natural ventilation based on proper engineering analysis. To undergo a proper analysis, the GHP Civil Engineering team must calculate the total emission of a particular harmful emission particle and determine the required airflow to dilute that particle in the tunnel for the given number of vehicles passing through. The team must also calculate the air entering the tunnel through natural means by both the natural air pressure or wind as well as airflow generated by the stack effect. The airflow from the stack effect is considered negligible due to the lack of sufficient temperature or elevation changes throughout the tunnel.

1.2. Summary of Results

The solutions recorded below display a required airflow of 3.85 m^3/s and a natural airflow of 98.98 m^3/s. The requirement for mechanical ventilation dictates that mechanical ventilation shall only be required if the required airflow is less than the airflow provided through natural means. Therefore, as 88.53 m^3/s > 3.85 m^3/s, the proposed tunnel design does not require the use of mechanical ventilation.

1.3. Input Requirements

- Number of personal vehicles
- Number of commercial vehicles
- Emission Factor of personal vehicles
- Emission Factor of Commercial Vehicles
- Total Emission of cars through the tunnel
- Admissible concentration of CO in the air
- Concentration of natural CO in the air
- Density of air
- Width of tunnel opening
- Height of tunnel opening
- Velocity of wind
- Drag coefficient for each vehicle type
- Cross-section area for each vehicle type
- Velocity of vehicles

1.4. Assumptions

6.3.3.1 - Assume the roadway has a speed limit of 55 mph, and that all vehicles are maintaining a proper speed. According to the FHWA, the statutory speed limit for rural highways is 55 mph; it is reasonable to assume that all drivers will adhere to the posted speed limit on the roadway in question, barring unusual traffic or weather conditions.

6.3.3.2 - Assume the traffic flowing through the tunnel includes 2500 passenger vehicles and 500 commercial vehicles. This assumption is based on a conservative estimate that increases the rate at which O2 is emitted within the tunnel; this increased emission ensures that the tunnel's natural ventilation is sufficient for higher traffic demands.

1.5. Calculator

Determine the amount of airflow needed to dilute the CO released by vehicles entering the tunnel. This step is calculated using equations 1.1 - 1.3, which take into account the emission factor of both commercial and personal vehicles as well as the admissible concentration of CO in the tunnel minus the concentration naturally occurring in the atmosphere. This calculator yields a result displaying a required airflow of 2.31 m^3/s. Using equations:

$$\frac{P}{2400} \times EP = G_p \text{ (Eq. 1.1)}$$

$$\frac{C}{2400} \times EC = G_C \text{ (Eq. 1.2)}$$

$$\frac{G_t}{C_{adm} - C_{atm}} = Q_{required} \text{ (Eq. 1.3)}$$

Determine the wind pressure caused by naturally occurring winds outside and through the tunnel. This is displayed in equation 2.1, which utilizes the density of air and velocity of wind (determined by the average wind speed in North Georgia), yielding a pressure of 1.91 Pa.

Equations used:

$$\frac{1}{2} \times \rho_{air} \times (v_{wind})^2 = P_{wind} \text{ (Eq. 2.1)}$$

Determine the pressure caused by the Piston Effect through the tunnel. This is showcased in equations 2.4 and 2.5, calculating the pressure caused by personal and commercial vehicles, respectively. This equation includes the blockage ratios calculated in equations 2.2 and 2.3, which determine what percent of the tunnel is being blocked and pushed by a single car. Equations 2.4 and 2.5 also utilize parameters such as the vehicle's velocity, the coefficient of drag for each given vehicle, the average number of cars passing per second, and the density of air. Solving each equation yields a total pressure generated by the piston effect of 7.41 Pa.

Equations used:

$$\frac{Ca_p}{CSA} = b_p \text{ (Eq. 2.4)}$$

$$\frac{Ca_c}{CSA} = b_c \quad \text{(Eq. 2.3)}$$

$$\frac{p}{2400} \times \frac{1}{2} \times \rho_{air} \times cd_p \times b_p \times (v_{vehicle})^2 = Pp_{piston} \quad \text{(Eq. 2.4)}$$

$$\frac{c}{2400} \times \frac{1}{2} \times \rho_{air} \times cd_c \times b_c \times (v_{vehicle})^2 = Pc_{piston} \quad \text{(Eq. 2.5)}$$

Use the total pressure (total pressure of piston effect plus the wind pressure) to calculate the average velocity of air traveling through the tunnel. This step utilized a derivation summarized in equation 3.1, in which the total pressure through the tunnel is set equal to the pressure lost, which can then be utilized with the parameters of the friction factor, height, and width of the tunnel cross section to isolate the average velocity. Calculating this equation yields a result dictating that the average wind speed through the tunnel is 0.52 m/s.
Equations used:

$$\sqrt{(2(P_t) \times (ff(\frac{t_h}{t_l}))} = v_{natural} \quad \text{(Eq. 3.1)}$$

Determine the volume of air being pushed through the tunnel. This step uses equation 3.2, which utilizes the previously calculated velocity of air through the tunnel and multiplies it by the tunnel cross section. Solving this equation shows that the airflow produced from natural ventilation moves a volume of 98.98 m^3/s.
Equations used:

$$V_{natural} \times CSA = Q_p \quad \text{(Eq. 3.2)}$$

1.6. Software Usage: Calculations were done using Microsoft Excel

1.7. Referenced Codes and Standards

National Fire Protection Association publication NFPA 502 Chapter 11.1.1 was referenced to determine the justification for the need or lack thereof for mechanical ventilation.

World Road Association publication 30199,2019R02-EN Chapter 2.2 was used to derive the formulas needed to determine approximate wind flow required to dilute the release of toxic emissions.

Appendix D - Electrical Calculations

The Calculations provided in this Appendix display the justification and requirements for key design features.

1. Electrical Lighting

1.1. Purpose

The purpose of the electrical lighting calculations is to determine the total amount of electricity needed while ensuring that it meets the tunnel lighting standards. These calculations establish the lighting power, safety lighting, and auxiliary electrical loads. The final total required power can be used for the electrical design, generator selection, maintenance, and future tunnel emergency repairs.

1.2. Summary of Results

While there is no lighting required for vehicles in the tunnel, the pedestrian lighting system required a minimum of 10 ft lightposts spaced every 10 ft, resulting in 12 lighting fixtures in a baseline lighting load of 1,518 W. The emergency and safety lighting system added 1,150 W with 480 W for auxiliary electrical equipment. No mechanical ventilation load is required as fans are unnecessary for tunnels of this length.

The Total Required Tunnel Electrical Power (PRT) turned out to be: PRT = 3,148 W

This electrical load took into consideration electrical circuit protection, emergency power, and future maintenance of the tunnel lighting system.

1.3. Input Requirements

The following inputs were used in support of this calculation:

1.3.1. Tunnel geometry used for the lighting calculations is as follows:

- Tunnel length: 60 ft
- Tunnel width: 48 ft
- Tunnel floor area: 2,880 ft² (267.56064 m²)

1.3.2. Lighting was installed one fixture every 10 ft along each pedestrian sidewalk:

- 6 fixtures per side
- 12 total lighting fixtures

1.3.3. Lighting fixtures are:

- Maximum fixture power: 110 W
- Minimum fixture power: 70 W
- Ballast factor: 15%
- Ballast-adjusted maximum fixture power: 126.5 W
- Ballast-adjusted minimum fixture power: 80.5 W

1.4. Assumptions

1.4.1. Based on the American Association of State Highway and Transportation Officials (AASHTO) Roadway Lighting Design Guide, according to the Georgia Department of Transportation (GDOT), vehicular tunnel lighting is not required for tunnels less than 80 ft. Due to the tunnel being developed through this project being 60 ft, no supplemental lighting calculations are needed for vehicle travel lanes.

1.4.2. As there are pedestrian walkways through this tunnel, pedestrian lighting is needed. In accordance with the Maryland State Highway Administration Bicycle and Pedestrian Design Guidelines, pedestrian tunnels are required to have a minimum lamp every 10 ft lamps to improve pedestrian safety.

1.5. Calculations

Total Required Tunnel Power:

$$PRT = PBL + PSL + PV + P_{aux} \text{ (Eq. 1)}$$

$$PBL = Number\ of\ Fixtures \times Fixture\ Wattage \times Ballast\ Factor \text{ (Eq. 2)}$$

$$LPD = Total\ Lighting\ Power \div Floor\ Area \text{ (Eq.3)}$$

$$PSL = P_{th} + P_{tr} + P_{ext} \text{ (Eq. 4)}$$

$$P_{aux} = Secondary\ Pump + Generator \text{ (Eq. 5)}$$

Step 1. Determine the baseline lighting requirements for the tunnel. As the tunnel is 60 ft long, the lighting needs to be spaced every 10 ft. This allows 6 fixtures per side and 12 total fixtures. The maximum fixture is 110 W, and the minimum is 70 W, adjusted using a 15% ballast factor, creating 126.5 W and 80.5 W.

$$60\,ft \div 10\,ft = 6\,fixtures\ per\ side$$
$$110\,W \times 1.15 = 126.5\,W$$
$$70\,W \times 1.15 = 80.5\,W$$

Step 2. Calculate the baseline lighting power (PBL) using equation 1, which is the total number of fixture wattages.
The baseline lighting power was calculated as:

$$PBL = 12\,V \times 126.5\,A = 1,518\,W$$
$$1,518\,W \div 267.56\,m^2 = 5.67\,W/m^2$$

Step 3. Determine the support lighting load (PSL) by calculating the electrical demand of the tunnel threshold lighting (Pth), transition lighting (Ptr), and the exit lighting (Pext). equations 1 through 4 calculate to 506 W, 322 W, and 322 W, with a total of 1,150 W.

$$P_{th} = 4\,V \times 126.5\,A = 506\ W$$
$$P_{tr} = 4\,V \times 80.5\,A = 322\,W$$
$$P_{ext} = 4\,V \times 80.5\ A = 322\ W$$

Step 4. Determine the final electrical loads. As the tunnel is less than 80 ft, supplemental vehicular ventilation was not needed, causing the ventilation load of PV = 0 W. The auxiliary electrical load (Paux) of the pump and a generator was a total of 480 W.

Then, PSL = 506 W + 322 W+ 322 W= 1,150 W. The auxiliary load was Paux = 480 W without any fans for ventilation PV = 0 W.

Step 5. Calculating the total required electrical power in the tunnel using equation 1. The total required electrical power in the tunnel calculated to be:

$$PRT = 1,518\,W + 1,150\,W + 0\,W + 480\,W = 3,148W \text{ (Eq. 1)}$$

1.6. Referenced Codes and Standards

Vehicular Tunnel Lighting AASHTO Roadway Lighting Design Guide (Referenced by the Georgia Department of Transportation) states that there is no supplemental lighting required for vehicular tunnels less than 80 ft long. However, the codes from the Maryland SHA Bicycle and Pedestrian Design Guidelines require pedestrian tunnels to provide a minimum illumination level of 10 ft candles.

Appendix E - Drainage Calculations

1. Drainage Calculations

1.1. Purpose

The purpose of the stormwater calculations is to ensure that the correct pipe sizing is used to accommodate the rainwater intensity and maintain proper flow in this mountain region. Additionally, these calculations ensure that trench drains and catch basins are properly sized. This sizing is based upon Georgia state and federal codes regarding stormwater management. Furthermore, these calculations ensure that pedestrians and drivers will not face dangerous situations regarding the inability for rainwater to flow.

1.2. Summary of Results

The rational method determined a value of 75.59 gpm, which is the peak surface runoff rate if a weather event were to occur in this area. Manning's equation had a value of 1.85 cfs; when compared using an 8 in pipe, it was determined that the pipe could not handle more than 1.85 cfs- which is significantly more than 75.59 gpm, permitting an 8 in pipe. Additionally, the wet well volume of 188.98 gpm is the required storage value of a wet well, permitting at least a 4 ft diameter well. Finally, the total dynamic head value of 10.64 ft is the total pressure the centrifugal pumps within the wet well must overcome to move fluid up and out of the well. Therefore, the vertical distance is 10 ft; there needs to be enough additional energy present to overcome the 0.64 ft.

1.3. Input Requirements

All input requirements are listed in Table 4.

Table 4 - Drainage Inputs

Variables		**Value**	**Units**
Q	**peak runoff rate**		**gpm**
C	runoff coefficient	0.91	
i	rainfall intensity	2.80	in/hr
A	drainage area	0.07	acres
Q	**flow rate**		**cfs**
n	pipe material coefficient (HDPE)	0.01	
A	pipe diameter	0.35	ft
S	pipe slope	0.02	
Rh	hydraulic radius	0.17	ft
hs	vertical distance	10.00	ft

Variables		**Value**	**Units**
hf	friction head loss	0.64	ft
TDH	**total dynamic head**		**ft**
Q	flow rate	75.00	gpm
V	**active storage**		**gpm**
theta	min cycle time	10.00	min

1.4. Assumptions

- Based upon past geotechnical surveys, it is under the assumption that there is a current stormwater main that runs parallel to the surface of the road.
- However, this tunnel lacks stormwater vents that allow for proper flow, avoiding pooling in the street during large events.
- Furthermore, it is assumed that the rock of the mountain is not too thick for the well water to not have the ability to be pumped out of the side of it.
- It is also under the assumption that the design requires modeling a system that reflects a 50-year storm coefficient.
- Additionally, this roadway uses the rainwater and climate qualities (annual rainfall of ~50 in) of the mountainous region of "Duluth, Georgia."

1.5. Equations and Calculations

- Calculate the rational method, dictating the peak flow rate from a single storm event that could flood this 60 ft tunnel.
- Calculate Manning's Equation, determining how much water a sloped pipe can carry using solely gravity.
- Calculate wet well volume; this determines how large the wet well needs to be to accommodate the pump size.
- Calculate total dynamic head; this portrays the necessary pump output to lift water out of the 10-ft sump pit and be discharged out of the 3 in, 20 ft PVC discharge pipe.
  - $Q = C \times I \times A$ (Eq. 1)
  - $Q = \frac{1.49}{n} \times A \times \frac{Rh^2}{3} \times \sqrt{S}$ (Eq. 2)
  - $V = \frac{Q\theta}{4}$ (Eq. 3)
  - $TDH = h_s + h_f$ (Eq. 4)

1.6. Software Usage: Excel was used to perform the drainage and sump system calculations.

1.7. Referenced Codes and Standards

2018 International Plumbing Code:

- Section 1101 was referenced for basic understanding of prohibited materials and methods of stormwater design
- Section 1102 was referenced for material usage
- Section 1113 was referenced for well and sump system code
- Appendix E was referenced for pipe sizing equations and tables

Georgia Department of Transportation:

- Section 5.2.2 was referenced for Georgia storm frequencies and rainwater intensities
- Section 5.3.1 was referenced for pipe sizing based off rain frequencies and constant

Appendix F - Drawings and Designs

This floorplan depicts the two-lane highway, along with a bike path and sidewalk that properly follows safety and ADA standards. Additionally, represented on this floorplan are various buffer zones that separate bike lanes from roadways and sidewalks. On the left, there is a small concrete strip that is not characterized as a sidewalk; however, if proper maintenance needs to be done, this can be completed along this concrete strip. Additionally, a utility hut is depicted that will be shown outside of the mountain for clarity that will house electrical components to the system. This does not represent the final location of the utility hut with regards to the thickness of the mountain.

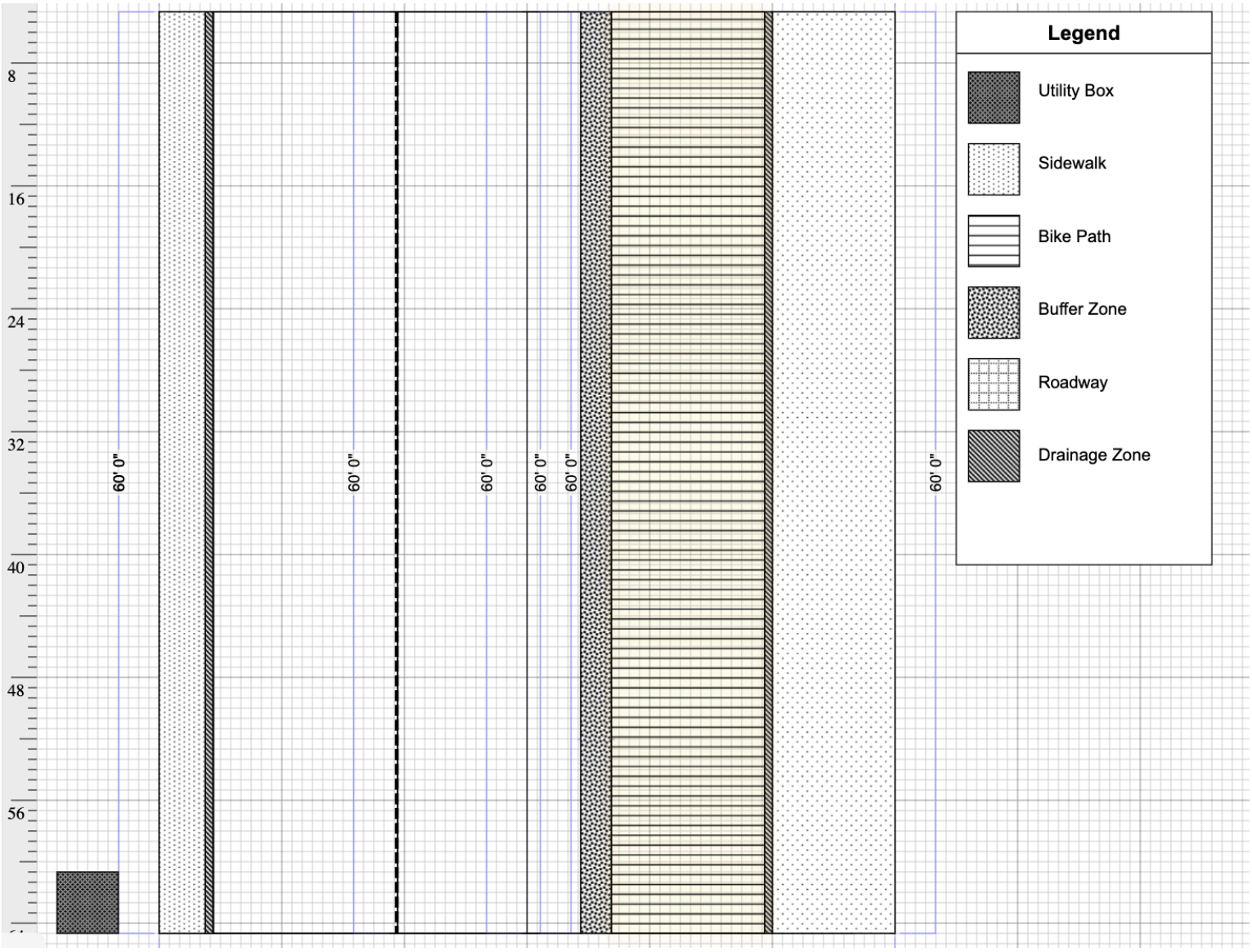


Figure 3 - Floorplan

This diagram mirrors the desirable piping and instrumentation system to accommodate proper lighting, drainage, and auxiliary systems that prioritize human safety. The source of electricity will be provided by the power line (15 kV), which is transmitted to the utility rooms. On the other hand, drainage inlets such as trench drains and catch basin lines prevent flooding on the roadway. The electrical and drainage systems primarily interact when it comes to the wet well system. This is because there are two VFDs running a primary and back-up pump (480 V) that push water up and out of the side of the mountain. These VFDs are located in the utility room, which is also home to the junction box. This box contains the wiring for several sensors located in the wet well that notify those when issues occur, such as a pump failure. The utility room also houses the lighting control panel, which powers the lighting system (120 V) inside the tunnel. Finally, auxiliary systems such as a back-up pump and generator are displayed to ensure proper measures are taken if one system fails.

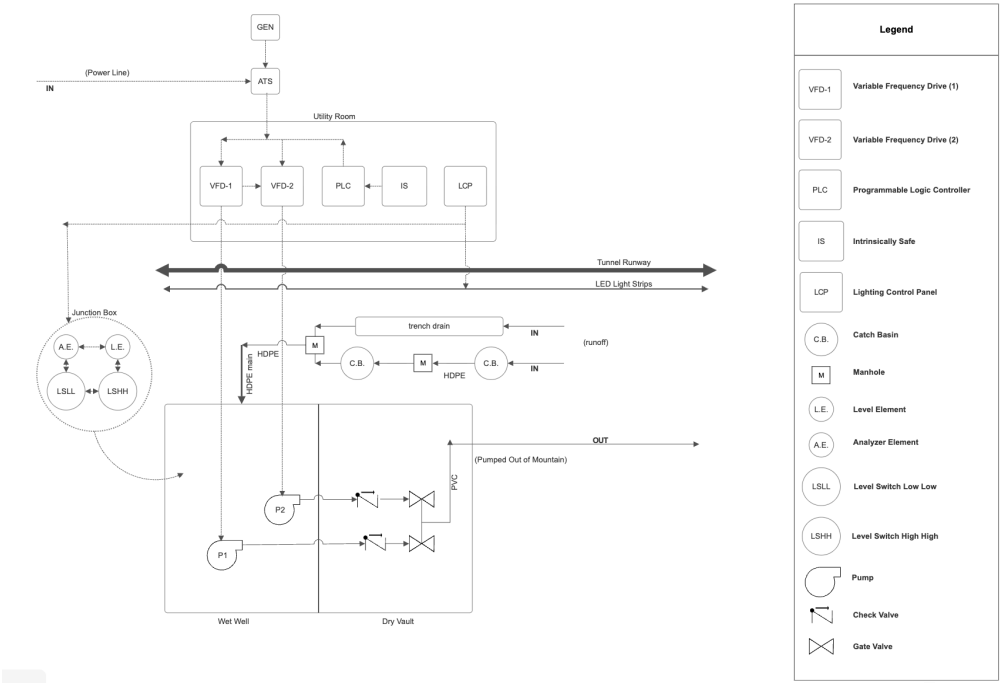


Figure 4 - Piping and Instrumentation Diagram

The elevation plan portrays the proper sizing of the tunnel. Primarily incorporating the various layers to the road while maintaining the proper concrete wall thickness. Additionally, this elevation plan includes a legend that describes piping and lighting symbolism. The piping system begins with a catch basin and trench drainage that ties together and finally connects to the wet well/dry vault where it is pumped out of the side of the mountain. This does not represent the final location of the utility hut with regards to the thickness of the mountain.

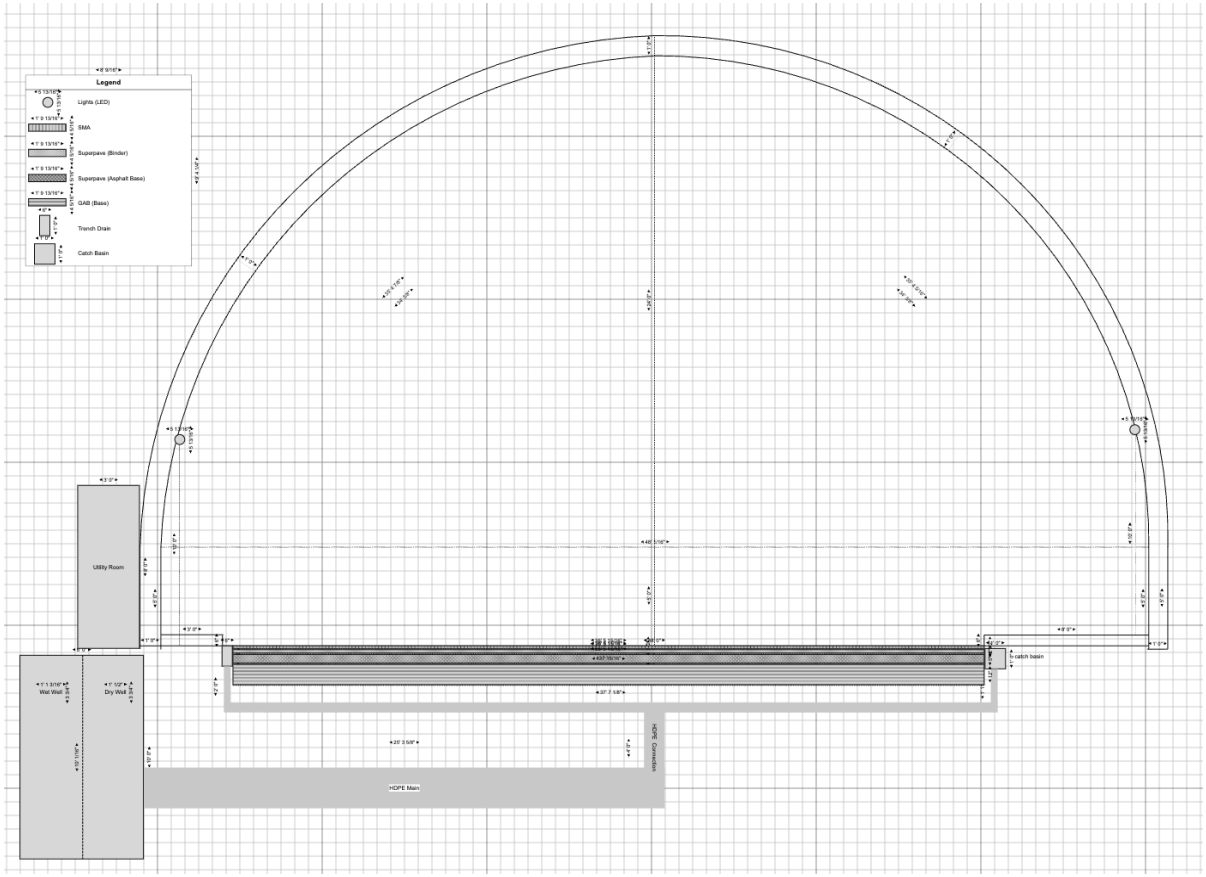


Figure 5 - Elevation

This electrical system represents the interaction of all of the different electrical components while also incorporating auxiliary lines in case of system failure. The primary inlet of electricity comes from the power line (15 kV) that powers the overall load bus system. This system uses VFDs to directly fuel the primary and back-up pump, while using a step-down transformer to power the lighting system. This system contains auxiliary components such as a generator. Additionally, the junction box with many of the pump sensors is represented with a connection to the PLC/IS components.

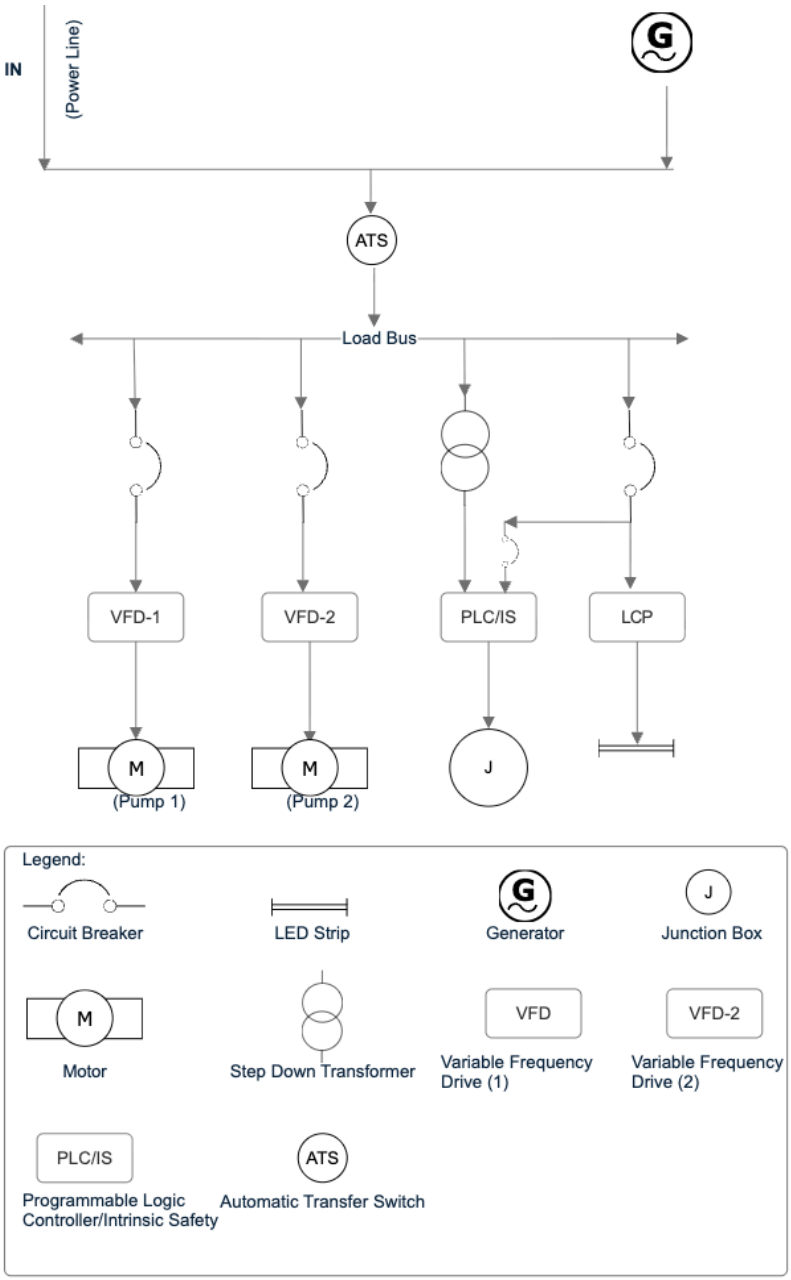


Figure 6 - Electrical One Line

This diagram depicts a 3D representation of the tunnel stormwater drainage system. On the left side of the tunnel two catch basins are connected by an 8 in HDPE pipe that drops down and connects to the piping that drops out of the rightward trench drain. These pipes connect at a T and drop down with 12 in HDPE, connecting to the already present 18 in HDPE stormwater main, which relays an outflow into a drilled wet well with a sump pump system. Additionally, manholes are added between the catch basins and connected to the storm main. Finally, this well has a 3 in PVC pipe that allows the pumped water to flow up and out of the mountain.

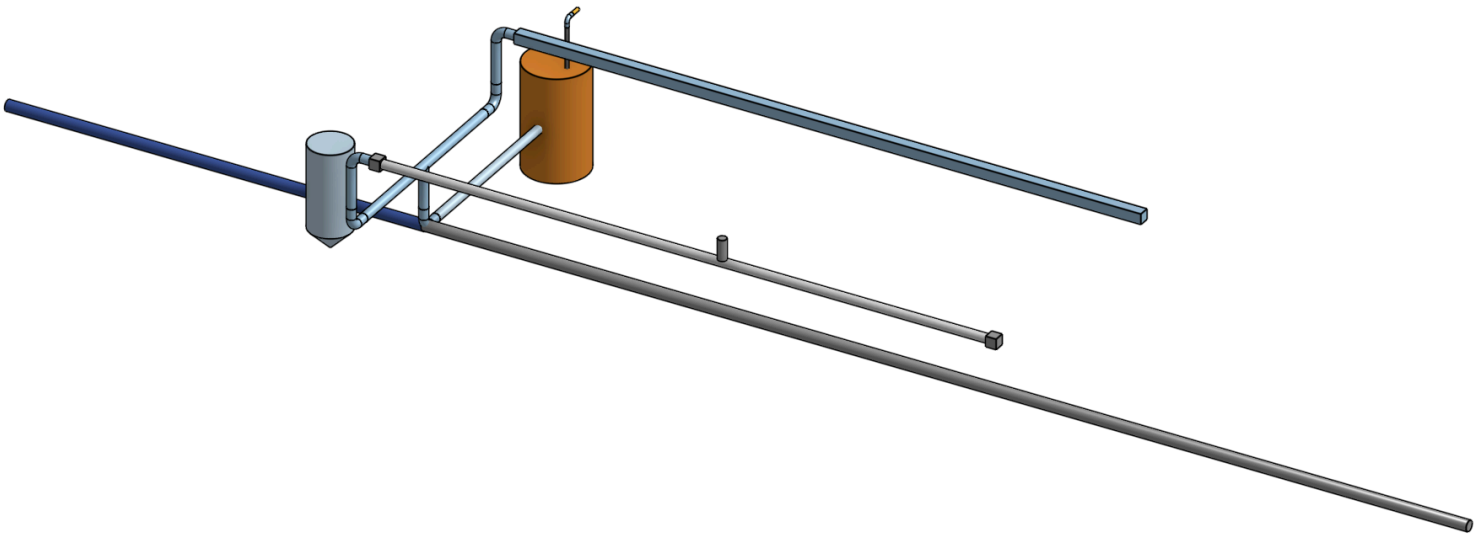


Figure 7 - Isometric

Appendix G - Bill of Materials

Table 5 - Bill of Materials for Real World Construction

Phase	Description	Qty	Item Number	Unit Price	Cost
Preconstruction - Project management	**Organize scheduling and communication with the client and ensure proper construction in accordance with the contract.**	**N/A**	**0**	**N/A**	**$1,463,819.32**
Project manager (s)	Control the financial and contractual requirements, developing proper scheduling	1	0.1	1-5% of project cost	$560,113.72
Project Superintendent (s)	Manage crew on the field ensuring proper production and site safety	8	0.2	$50.43 per hour	$903,705.60
Land Surveying	**Identify materials surrounding structures, ensuring stability and preventing collapse, also ensuring any hidden preexisting utilities or structural hazards are identified and avoided.**	**N/A**	**1**	**Total cost for items labeled 1.xx**	**$54,754.48**
Survey Crew/Company	Workers used in identifying materials surrounding structures, ensuring stability and preventing collapse, also ensuring any hidden preexisting utilities or structural hazards are identified and avoided.	1	1.3	$1500 per day	$30,000.00
GPS Survey Equipment	Used to determine exact coordinates and distance measurements of the job site	1	1.4	$2500 per unit	$2,500.00
Terrestrial lidar scanner	Used to capture point clouds displaying structures and characteristics of the job site	1	1.5	$19000 per unit	$19,000.00
Laser Distance Meters	Used to capture measurements	1	1.6	$200 per unit	$200.00

	of boundaries and property dimensions				
Crack Gauges	Used to measure and track the widening and expansion of cracks within the existing structures	3	1.7	$350 per kit	$1,050.00
Utility Locating Equipment	Used to highlight and mark utility lines such as electricity and water utilities	1	1.8	$1600 per month to rent	$1,600.00
Geotechnical Survey	**Evaluates the soil, rock, and water conditions before construction**		**2**	**Total cost for items labeled 2.xx**	**$137,244.48**
Geotechnical Engineering Crew	Workers used in evaluating the soil, rock, and water conditions before construction	Crew size 2	2.1	$275 per hour	$57,200.00
Drilling Crew	Operators using drilling rigs to extract rock cores for testing	Crew size 2	2.2	$27.84 per hour	$2,004.48
Laboratory Technicians	Technicians analyzing and providing material reports for rock core samples	Crew size 2	2.3	$250 per hour	$40,000.00
Drill Rig	Rig used to dig into and expose new rock	1	2.5	$8000 per week	$32,000.00
Borehole Camera	Cameras used to look into holes created to analyze rock layers and composition.	1	2.6	$3000	$3,000.00
Rock Coring Equipment	Special equipment used to create a cylindrical rock sample to be sent to testing	2	2.7	$380	$3,040.00
Existing tunnel inspection	**Assess current structural stability, locate potential developed hazards, and ensure safe excavation.**	**N/A**	**3**	**Total cost for items labeled 3.xx**	**$27,732.00**
Structural Engineer	Engineering crew analyzing	Crew size: 2	3.1	$200 per hour per engineer	$6,400.00

	the structure's ability to handle the stresses during the excavation process and ensure the new structure can withstand proper forces while being up to code				
Tunnel Inspection Crew	Inspect the preexisting tunnels' condition and ability to be safely worked on	1	3.2	$301 per hour per crew	$4,816.00
Lift Platforms	Platform able to be raised and lowered, allowing inspectors to closely and safely analyze the condition of higher tunnel regions	2	3.3	$379 per day source	$1,516.00
Mobile Scaffolding	Longer lasting stationary platforms used in high elevation tunnel inspection	3	3.4	$1200 per kit	$3,600.00
Schmidt Hammer	Non-destructive concrete testing tool to estimate the compressive strength of concrete liners	2	3.6	$700 per hammer	$1,400.00
Ultrasonic Concrete Tester	Further analysis tool for testing concrete condition and strength of existing lining	1	3.7	$10000	$10,000.00
Traffic control and site setup	**Offer essential detours and control proper traffic safety to avoid disruption during construction.**	**N/A**	**4**	**Total cost for items labeled 4.xx**	**$115,400.00**
Traffic Control Crew	Crew used to direct the flow of traffic and ensure safe travel conditions during work time.	1	4.1	$167.50 per hour	$26,800.00
Traffic Barriers	Used to divide the road and separate workers from flowing traffic	6	4.2	$1000 per ten foot block	$6,000.00
Arrow Boards/Signage	Signs utilized to warn	2	4.3	$1250 per week	$70,000.00

	incoming drivers of delays and lane closures				
Portable Lighting	Lighting used to illuminate the inside of the tunnel to increase safe working conditions.	3	4.4	$225 per week	$12,600.00
Excavation preparation	Assemble and acquire all required vehicles and materials needed for the excavation process.	N/A	4.5	N/A	$705,627.20
Blasting Engineer	Used to plan, design, and oversee controlled explosions used for excavation	Crew size 2	4.6	$48.92 per hour	$7,827.20
Hydraulic Excavators	Large excavators used to clear debris and rubble from controlled explosions, as well as excavated debris from drainage excavations	1	4.7	$2000 per week	$64,000.00
Scaling Equipment	Heavy machinery used to scrape away and break loose rock from the tunnel roof and walls.	1	4.8	$3000 per week	$96,000.00
Rock Drills	Used to drill precise holes to place explosives in for controlled explosion	1	4.9	$10,000 per week	$20,000.00
Loaders	Used for “mucking” clearing smaller rubble from excavation processes	2	4.10	$2500 per week	$160,000.00
Dump Trucks	Used to haul collected debris to controlled disposal sites	2	4.11	$4500 per week	$288,000.00
Ventilation Fans	Used to pump clean air and clear dust from the tunnel during the excavation process	4	4.12	$500 per week	$32,000.00
Lubricants	Used to reduce friction and wear in equipment and reduce	3	4.13	$600 per drum	$1,800.00

	heat produced during excavation				
Dust Suppression Equipment	Used to clear the air during excavation protecting works respiratory systems and sight.	2	4.14	$6000 per month	$36,000.00
Tunnel widening excavation	**The tunnel is expanded using the drill and blast method, essential for adding space for additional bicycle and pedestrian walking lanes.**	**N/A**	**5**	**Total cost for items labeled 5.xx**	**$1,532,222.40**
Blasting Crew	Crew performing the labor of placing explosives and detonation	Crew size 4	5.1	$300 per hour	$192,000.00
Drill Operators	Operators using drilling equipment to create the holes explosives are placed in	2	5.2	$27.84 per hour	$8,908.80
Excavator Operators	Operators using the excavators to remove large debris	2	5.3	$29.57 per hour	$75,699.20
Loader Operators	Operators monitoring and using loaders to collect and move smaller debris	2	5.4	$19.80 per hour	$50,688.00
Haul Truck Drivers	Drivers moving in and out of the tunnel and overall site, moving excavated materials.	2	5.5	$34.24 per hour	$87,654.40
Scaling Crew	Crew focusing on the removal of loose rock on the tunnel interior	1	5.6	$3000 per day	$540,000.00
General Laborers	General construction laborers carrying out overall site tasks as dictated by the superintendent	15	5.7	$22.41 per hour per hour	$430,272.00
Generators	Equipment used to produce electricity needed within the tunnel while power lines are	2	5.8	$1500 per week	$96,000.00

	disconnected				
Lighting Towers	Towers used to illuminate the working environment within the tunnel	3	5.9	$350 per week	$44,800.00
Explosives	The actual explosives being detonated within drill holes to break free and clear out rock within the tunnel	4	5.10	$1100 per ton	$4,400.00
Hydraulic Oil	Used in the heavy equipment to create a high-pressure system and high strength for excavators, rock splitters, etc	3	5.11	$600 per drum	$1,800.00
Temporary ground support	**Supports added during the excavation process to prevent tunnel collapse**	**N/A**	**6**	**Total cost for items labeled 6.xx**	**$12,100.00**
Rock bolts	Reinforce fractures rock masses providing support before permanent lining is supplied	100	6.1	$30 per bar	$3,000.00
Grout	Used to fill in empty spaces where undesired rock fell, as well as to prevent potential floodwater from spilling into the site	5	6.2	$20 per cubic foot	$100.00
Monitoring Equipment	Used to track ground deformation, stress distribution, and groundwater changes within the site in real time	1	6.3	$1500 per month	$9,000.00
Permanent Structural Concrete Lining & Shell	**Covers the bulk raw materials, internal steel reinforcement, and structural framing assets needed to physically build the permanent lining shell.**	**N/A**	**7**	**Total cost for items labeled 7.xx**	**$102,622.50**

High-Strength Concrete	Forms permanent 12-inch structural lining	190	7.1	$165 per cubic yard	$31,350.00
#8 Grade 60 Deformed Steel Rebar	Acts as heavy main reinforcement cage running along internal geometry	16.5	7.2	$1250 per ton	$20,625.00
Structural Chopped Steel Fibers	Fibers are mixed directly into the concrete mix to reinforce the arch	3,150	7.3	$1.65 per LB	$5,197.50
Modular Horseshoe Framework	Custom steel framework rental, delivery and stripping	1	7.4	$35,000	$35,000.00
Non-shrink Cementitious Structural Grout	Contact grouting between concrete lining and the raw rock face	475	7.5	$22 per bag (50 lb)	$10,450.00
Labor (tunnel lining)	**Tracks the field hours and burdened commercial wage rates for the specialized crews required to assemble the steel cages and manage the formwork operations.**	**N/A**	**8**	**Total cost for items labeled 8.xx**	**$18,720.00**
Certified Structural Steel Welders	Fabricate and tie the main inner lining reinforcement cages	Crew of 2	8.1	$65 per hr	$6,240.00
Concrete Formwork & Slick Line Operators	Assemble structural horseshoe framing and handle concrete pumps	Crew of 3	8.2	$52 per hr	$12,480.00
Machinery / Equipment	Outlines the logistical transport fees, fuel reserves, and weekly rental rates for the equipment fleet required during active construction cycles.	N/A	8.3	N/A	$23,607.50
Heavy-Duty Concrete Slick-Line Pump Truck	High-pressure structural pump to inject concrete into the modular horseshoe arch forms	1 (Weekly Rental)	8.4	$3,850	$3,850.00

Concrete Transit Mixer Truck (10 CY)	Maintains batch uniformity from the staging yard into the slick-line pump	2 (Weekly Rental)	8.5	$1,450	$1,450.00
Telehandler (10k lbs Capacity)	Staging, lifting, and aligning the modular steel formwork panels and heavy rebar bundles	1 (Weekly Rental)	8.6	$1,200	$3,600.00
Industrial Concrete Vibrators	Handheld high-frequency consolidators to eliminate air pockets	3 (Weekly Rental)	8.7	$115	$115.00
Pneumatic Grout Injection Pump	Low-volume, high-pressure pump to execute contact grouting between the raw rock face and the lining	1 (Weekly Rental)	8.8	$650	$650.00
Mobile Diesel Generator Rig (150 kW)	Supplies continuous primary tier-4 power for slick-lines, vibrators, lighting, and site tools	1 (Weekly Rental)	8.9	$1,100	$3,300.00
Heavy Equipment Mobilization	Flat-bed freight transit of pump trucks, telehandlers, and generator units to site	1	8.10	$4,500	$ 4,500.00
Ultra-Low Sulfur Diesel Fuel	Fuel consumption allowance across all active on-site equipment engines	1,350	8.11	$4.55 per gallon	$6,142.50
Road Pavement and Sidewalk Basis			**9**	**Total cost for items labeled 9.xx**	**$26,247.00**
Graded Aggregate Base (GAB)	A compacted mixture of crushed rock to serve as the primary foundation of the roadway	12” thick layer over 2160 sq ft area	9.1	$75 per ton	$8,100.00
Superpave Binder	Liquid bitumen material used to bind asphalt layers together	2” thick layer over 2160 sq ft area (360 ft^3)	9.2	$120 per ton	$3,200.00
Superpave Asphalt Base	Asphalt and rock structural layer to withstand large	5” thick layer over 2160 sq ft area (900 ft^3)	9.3	$100 per ton	$6,667.00

	vehicle loads without road deformation				
Stone Matrix Asphalt (SMA): GP 2 only, Poly-Mod. & H. Lime	Durable asphalt surface mixture to serve as the top layer of the road	1.5” thick layer over 2160 sq ft area (270 ft^3)	9.4	$150 per ton	$3,000.00
Portland Cement Concrete for Sidewalk	Strong concrete mix to serve as the basis of a sidewalk	4” thick layer over 660 sq ft area (220 ft^3)	9.5	$8 per sq ft (4” thick layer)	$5,280.00
Road Add-Ons					$1,280.00
Flexible Delineators	Reflective posts to mark the buffer zone between the vehicle lanes and bike lane	7 markers (placed every 10 ft along buffer zone)	2.6	$40 per post	$280.00
Thermoplastic Traffic Paint	Reflective paint used to mark lane boundaries	~500ft of 4-inch wide lines	2.7	$2 per linear foot of 4” wide stripe	$1,000.00
Machinery			**10**	**Total cost for items labeled 10.xx**	**$6,100.00**
Skid Steer	Small construction machines able to be equipped with buckets, land planes, rollers, spreader boxes, and more to pave roads	2	10.1	$3000 each per month	$6,000.00
Line Striping Machine	Small machine to spray paint lines on the road	1	10.2	$100 per day	$100.00
Labor Crews			**11**	**Total cost for items labeled 11.xx**	**$7,000.00**
Skid Steer Crew	Operate the skid steer for construction of the road	Crew of 2-3	11.1	$500 per day	$2,500.00
Sidewalk Construction Crew	Build and manage construction of sidewalks and concrete curbs	Crew of 5	11.2	$1500 per day	$3,000.00
Road Painting Crew	Operate the line striping machine to paint the roads and	Crew of 2-3	11.3	$1500 per day	$1,500.00

	add flexible delineators to the buffer zone				
Electrical Installation			**12**	**Total cost for items labeled 12.xx**	**$16,659.30**
Power distribution	Main service disconnect panelboard	1	12.1	$500 each unit	$500.00
Power distribution	Enclosed NEMA 3R weatherproof safety housings	1	12.2	$330 each unit	$330.00
Power distribution	Molded case circuit breakers	1	12.3	$390 each unit	$390.00
Conduit system	Rigid intermediate metal conduit (IMC)	80	12.4	$6.71 per ft	$536.80
Conduit system	PVC-coated rigid conduit paths	20	12.5	$34.50 per ft	$690.00
Conduit system	Liquid-tight flexible metallic conduit connections	10	12.6	$5.35 per ft	$53.50
Industrial wiring	Stranded copper building wire (THHN/THWN-2 insulation cores, rated for wet/damp infrastructure environments)	400	12.7	$8.25 per ft	$3,300.00
System enclosures	Surface-mounted NEMA 4X stainless steel junction boxes	400	12.8	$8.25 per ft	$3,300.00
System enclosures	Heavy-duty grounding terminal blocks	2	12.9	$285 each unit	$564.00
System enclosures	Copper grounding rods and grounding clamps	1	12.10	$195 per lot	$195.00
Lights	Required lightbulbs in electrical system	12	12.11	$150 per bulb	$1,800.00
Installation labor	Workers required for installing all lights	2 workers in 5 days	12.12	$500 people/day	$5,000.00

Drainage System Installment			**13**	**Total cost for items labeled 13.xx**	**$43,068.99**
8” HDPE Pipe	Basic flow pipe that connects between catch basins and trench drains.	126’, 8” HDPE	13.1	$7.5 per ft	$945
12” HDPE Pipe	Flow pipe that connects between left trench drains and right catch basin systems.	4’, 12” HDPE	13.2	$10 per ft	$40
18” HDPE Pipe	Connection to stormwater main.	30’, 18” HDPE	13.3	$10 per ft	$300
8” HDPE 90	Drops the basic flow down into the connection pipe.	4x HDPE 90s	13.4	$175 per fitting	$900
HDPE T	Drops connection pipe down to stormwater main.	1x 8” by 8” by 12” HDPE T	13.5	$475 per fitting	$475
HDPE Lateral T	Lateral T that connects current stormwater main to connection pipe	1x 18” by 18” by 12” HDPE Lateral T	13.6	$1,207.08 per fitting	$1,207.08
18” HDPE 90	90 that connects stormwater line to filtration system	1x 18” HDPE 90	13.7	$219.99 per fitting	$219.99
HDPE Bonding Cement (XL)	Sealant for piping and fittings	1x lXL HDPE Bonding Cement	13.8	$567 per canister	$567
8” PVC Pipe	PVC that connects centrifugal pumps to exit out of mountain	15’, 8” PVC 90	13.9	$16.34 per foot	$245.17
8” PVC 90	90 officially exits mountain	1x 8” PVC 90	13.10	$183.25 per fitting	$183.25
Sediment Filter	Filter between storm main pipe and wet well	1x Sediment Filter	13.11	$661 per filter	$661
Commercial Catch Basin	Piping vents on right side of tunnel	2x Commercial Catch Basin	13.12	$340 per catch basin	$680

Manholes	Cleanouts between catch basins and connection between right and left side of tunnel	2x Manholes	13.13	$1,391.25 per manhole	$2,782.50
Wet Well/Dry Vault	Contains electrical sensors and two centrifugal pumps	1x Wet Well/Dry Vault Installation	13.14	$27,000 per wet well/dry vault	$27,000
8” Fiber Reinforced Pre-Slope Channel	Trench drains allocated on the left side of the road.	60’, 8” fiber reinforced pre-slope channel	13.15	$111.38 per ft	$6,683
Outlet End Caps	Outlet end caps between catch basins and HDPE piping	2x outlet end caps	13.16	$90.00	$180.00
Total Cost (minus project management)					$3,734,000
Total Cost					**$5,198,000**